

# Pranav Kumar Soma

SOFTWARE ENGINEER • MACHINE LEARNING • AI RESEARCH

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## EDUCATION

### M.S. Computer Science, UC San Diego

Dec 2026 (expected)

GPA 4.00 • Jacobs School of Engineering • M.S. in 1.25 years

### B.S. Computer Science, UC San Diego

Dec 2025

GPA 3.96 • Cum Laude, CSE Department Honors, Provost Honors (all quarters) • B.S. in 2 years

## EXPERIENCE

### Founding Software Engineer | Stealth Startup

Starting Sep 2026

Autonomous systems • part-time

- Building the **core operating system** for a hardware-agnostic **autonomy stack**: multi-agent coordination & trajectory planning across distributed edge nodes, real-time **sensor fusion**, low-latency **C2** dashboards, **hardware-in-the-loop** simulation

### Software Engineer Intern | ServiceNow

Summers 2025 & 2026

AI Voice Agents, IT Service Management • **Technical Skills Award** • top intern of 800+

- Cut **AI Voice Agent latency 45%** by re-architecting the NLU pipeline onto higher-capacity models across 8 intents; shipped to production and validated live with **Disney+**; delivered the AWS Lex V1→V2 migration ahead of schedule, unblocking every ITSM voice deployment before end-of-life. **Return offer; invited back for a second summer**

### AI Research Fellow & Agentic Systems Lead | Laboratory for Emerging Intelligence, UCSD

Apr 2025–Present

- Shipped a fine-tuned **AI Tutor** to **8,000 students across 3 universities** (UCSD, SDSU, Cal Poly Pomona) • **64% fewer TA hours, 87% higher engagement** in A/B tests; LoRA + SFT + RLHF over Bayesian Knowledge Tracing
- Built its reliability layer: live output monitoring plus an **LLM panel-of-judges** scoring every response into a per-student learner model that steers the next turn
- Lead the **Teradata-funded** agentic platform for 7,000+ users: custom **MCP servers** over UCSD APIs, deterministic fault-graceful workflow executor

### Graduate Researcher | Qualcomm Institute / Calit2 (HXI Lab)

Nov 2024–Dec 2025

- Reached diagnostic accuracy validated against **EMS triage protocols** at sub-second query latency, by fine-tuning local Qwen models (SFT) on the SD Supercomputer Center and grounding them in **GraphRAG** over the UMLS ontology
- Built **TB-scale** multimodal pipelines turning raw sensor/audio/video into structured health-database queries (LifeSaver); first-author lab poster

### Honors & M.S. Thesis Researcher | UCSD CSE • Computational Redistricting

Sep 2024–Present

- Built 3 districting generators (**XGBoost** + graph partitioning at **MAE 0.042**, Sequential Monte Carlo, greedy DP + Kernighan–Lin) over Census-2020 data at **158K–5.5M blocks/state**; reproduces official district counts for AL, CA, TX

## SELECTED PROJECTS

- ORCA • AI Music Production** (*Best Project, SDTC: 1 of 30 from 200+; SAIRS 2025*) • OpenCV + hybrid neural nets for source separation, CNN audio-to-score; **98% arrangement accuracy**. Flask, Docker, React
- Spay.LA** (*spay.la*) • MERN fundraising & legislative platform; **raised \$75K**; led 11 developers. Firebase/Google auth
- AgentSpace** • hireable expert agents over two-way **GraphRAG** on skill embeddings, served over **MCP + A2A**

## LEADERSHIP

**Co-Founder & President**, Real World AI Network (*2026–Present*) • founded **7,000+ student** federation across UCSD’s engineering, data-science & business schools driving AI research & entrepreneurship.

**Founding Board Member & VP External Affairs**, University AI Alliance • student-led frontier-tech network spanning MIT, Stanford, Caltech & Cornell; own external relationships with a16z speedrun, Bain Capital Ventures, Techstars & execs at Google and Meta.

**President**, CSE Society @ UCSD (200+ members, 21+ projects) • **Founder**, Innovate Research Group (30+ across UCLA, UCSB, UCSC) •

**Awards**: YC Startup School 2026, National Merit Finalist, USACO Silver.

## TECHNICAL SKILLS

**Languages**: Python, C++, C, Java, TypeScript/JavaScript, SQL, Swift, Kotlin, ARM-32 Assembly, Bash

**ML & AI**: PyTorch, Transformers, scikit-learn, XGBoost, OpenCV; LLM fine-tuning (LoRA, SFT, RLHF), RAG & GraphRAG, agents & MCP, LLM-as-judge evals, multimodal models, RL, NLP/NLU, sequential Monte Carlo, computer vision

**Infra & Data**: FastAPI, Flask, React, Node/Express, REST APIs; NumPy, Pandas, GeoPandas, pymetis, Parquet; MongoDB (vector), PostgreSQL; AWS (Lex, Lambda), Docker, Git, Linux, multi-GPU computing, SD Supercomputer Center

**Systems**: Distributed & real-time systems, system design, sensor fusion, embedded systems, ROS; ServiceNow (ITSM, CSM, SPM)